MODULE 01 · CASE STUDY 4 · PRE-READ FOR CLASS 4

Counting a Market That Did Not Exist Yet

In 2016 almost every rupee in India moved as cash. Three men left the best job in Indian e-commerce to bet that this was about to change.

READ THIS BEFORE CLASS

This case has more numbers in it than any case you have read so far, and none of them are hard. Every calculation here can be done on paper with a pen, and you should actually do them as you read rather than letting your eye slide past — that habit is most of what the next session is about.

Bring paper and a pen to class. As always: no frameworks, no definitions, nothing to memorise. A country, three men who quit their jobs, and a lot of arithmetic that anybody can check.

18–22	8	5
Minutes to read	Sections	Questions to bring

Zenith School of AI · Customer Discovery · Semester 1

THE CASE

Counting a Market That Did Not Exist Yet

PhonePe, 2015 to 2024 — the ₹40 you hand the auto driver

READING TIME: 18–22 minutes

HOW TO READ THIS CASE

Keep one question running the whole way through: **where did that number come from?** Some of the figures you are about to meet were counted by somebody. Some were estimated. Some were simply asserted, confidently, by a person who wanted you to believe them. A large part of becoming good at business is being able to tell which is which at a glance, and it is a skill you can practise on this page.

Do the arithmetic as you go. If a section multiplies two numbers together, multiply them yourself before you read the answer. It will take you an extra ninety seconds per section and it is the entire point of the exercise.

1 India, 2015: A Country That Ran on Cash

Start with the country as it actually was, because the whole case turns on how unpromising it looked.

In 2015, roughly 98% of consumer transactions in India happened in cash. Not most — almost all. The vegetable seller took cash, the auto driver took cash, the chemist took cash, the landlord took cash, salaries in large parts of the economy were paid in cash, and the total currency in circulation was around ₹17 lakh crore of physical notes moving through hands every day.

The digital alternatives all existed and none of them worked at scale.

WHAT EXISTED IN 2015	WHY IT WAS NOT THE ANSWER
600 million+ debit cards	The overwhelming majority were used for exactly one thing — withdrawing cash from an ATM.
About 25 million credit cards	In a country of 1.3 billion. A metro phenomenon for a thin professional class.
About 1.5 million card machines in shops	Sounds large until you divide it across some 60 million small businesses — and the machine cost the shopkeeper a monthly rental plus a percentage of every sale.
Net banking	A login, a password, a security question, adding a payee, a cooling period before that payee became active, and a one-time password that arrived when it felt like it.

Mobile wallets had been tried. Airtel Money, mChek, Oxigen and others launched around 2012, several with real money and real distribution behind them. They failed to become anything. In 2012 India had fewer than 100 million

smartphones, mobile data cost enough that people switched it off between uses, and every wallet was a closed loop — money inside Airtel Money could not reach a person on Oxigen.

So the picture in 2015 was: 1.3 billion people, almost all of their transactions in cash, roughly 200 million smartphones and rising fast, no usable digital connection between ordinary people's bank accounts, and a decade of failed attempts to build one. Depending on how you looked at it, this was either the least attractive market in Indian technology or the largest one.

2 The Unlock: What UPI Actually Did

In April 2016 the National Payments Corporation of India — a not-for-profit body set up by the Reserve Bank and a group of banks — launched the Unified Payments Interface. It went public that August.

UPI is not an app and not a company. It is a set of rules that lets any bank account talk to any other bank account instantly, through any app, at any hour. Four properties made it different from everything before it:

PROPERTY	WHAT IT MEANT
Interoperable	Money in your account at one bank could reach a person at any other bank, through whichever app either of you happened to use.
Real time, always on	Including Sunday at 2am.
A simple address	It replaced the account number and IFSC code with a virtual payment address, and later just a phone number — removing the most error-prone step in the whole process.
Free	For person-to-person transfers: no charge to the user, and no fee taken out of the payment.

> **HOLD THIS THOUGHT**
>
> Read that last property twice. A free rail is the fastest possible way to get a country to adopt something.
>
> It is also, by construction, a rail on which the payment itself earns nobody any money. Keep that in your head until section 6, where it comes back.

There is a second piece of timing that nobody in this story could have predicted. On the 8th of November 2016, three months after UPI went public, the Government of India announced that ₹500 and ₹1,000 notes — around 86% of the currency in circulation by value — would cease to be legal tender. Whatever one thinks of demonetisation as policy, its effect on the question in this case is not disputable: for several weeks a very large number of Indians who had never made a digital payment were required to find another way to pay for things.

3 Three Men Who Left the Best Job in Indian E-Commerce

In December 2015 — four months before UPI launched, and eleven months before demonetisation — Sameer Nigam, Rahul Chari and Burzin Engineer resigned from Flipkart to start PhonePe.

This was not an obvious move. Flipkart in 2015 was the most valuable private company in India and by some distance the most desirable place to work in Indian technology. Sameer Nigam was a Vice President of Engineering there; he had earlier founded a company called Mime360 which Flipkart acquired, so he was already someone who had built

and sold something. Rahul Chari had run engineering teams at Flipkart and before that in the United States. Burzin Engineer had spent years on large-scale systems reliability.

What they had was not a product idea, and this is the part people consistently get wrong when they retell the story. PhonePe did not invent digital payments, did not invent UPI, and did not have a technology the other apps lacked. What the three of them had was a reading of a market at a specific moment, and the conviction to act on it fourteen months before the evidence arrived.

The reading went roughly like this. The rail is about to exist, and it will be free, open and interoperable — which means the payment itself becomes a commodity, something nobody can charge for. When something becomes a commodity, the value moves to whoever owns the customer relationship on top of it. Smartphone growth and collapsing data prices mean the number of people who can use such a rail will roughly double in three years. And a country where almost every transaction is cash has, by definition, the largest unconverted payments market on earth.

WHEN	WHAT HAPPENED
Dec 2015	Nigam, Chari and Engineer resign from Flipkart. UPI does not exist yet.
April 2016	UPI launches. In the same month Flipkart acquires PhonePe — four months old, with no product shipped.
Aug 2016	PhonePe launches publicly, one of the first UPI apps in the country.
Nov 2016	Demonetisation.
2018	Walmart acquires Flipkart, and PhonePe comes with it.
Dec 2022	PhonePe is separated out and redomiciled to India.
2023	Raises at a valuation of around $12 billion.

4 Which Market Were They Actually Counting?

Here is where the case becomes useful rather than merely interesting. Ask what market PhonePe was counting in 2016 and there are three defensible answers, each producing a wildly different number.

The first answer: online payments. In 2016 Indian e-commerce was doing perhaps ₹1.5 lakh crore of goods sold per year, and most of that was paid for in cash on delivery anyway. Count the payments opportunity against online shopping and you get a modest business — a payments layer inside a category that was itself still small. On that reading, PhonePe is a feature of Flipkart and not much more.

The second answer: digital payments. Add up the card transactions, the net-banking transfers and the wallet top-ups. Larger than the first, but still a thin slice of a cash economy, and growing at the sedate pace those instruments had grown at for a decade.

The third answer: every transaction in the Indian economy. Not every online transaction. Every transaction — the ₹40 you hand the autorickshaw driver, the ₹7,000 you give your landlord, the ₹200 to the chemist, the ₹15,000 paid to a household help.

> **WORTH PAUSING ON**
>
> Before you read on, pick one of those three and write it down. Then write one sentence saying why the other two are wrong.
>
> You will be asked for your pick by a show of hands in the first fifteen minutes of class, and the room usually splits three ways. Having committed on paper beforehand is the difference between having an argument and having an opinion.

5 The Arithmetic Somebody Made Them Defend

A conviction is where this starts, not where it ends. Any investor in 2016 would have made the three founders build the number from the ground up, one line at a time, and every line is a claim you can check. Follow along on paper; the arithmetic is deliberately simple.

Line one: how many Indians could plausibly use this at all? A user needs a bank account and a smartphone with data. In 2016 India had well over 700 million bank accounts after the Jan Dhan expansion, but many were dormant, so call the number of adults with a live, usable account around 500 million. Smartphone users were roughly 250 to 300 million, and were forecast to reach 500 million by 2020. The overlap of the two — people with both a working account and a smartphone — was perhaps 200 million in 2016, heading towards 400 million within four years.

Line two: what fraction of them will actually use it regularly within three years? This is a judgement, and it is where honest people disagree. Say one in three of those eligible becomes a monthly-or-more user by 2019. That is roughly 100 million people.

Line three: how much does one of them send in a month? Start conservatively. Say ten transactions a month at an average value of ₹700 — a mix of small shop payments and larger transfers between people. That is ₹7,000 per user per month.

Now multiply, slowly.

STEP	NUMBER
100 million users × ₹7,000 a month	₹70,000 crore of payments a month
Across a year	roughly ₹8.4 lakh crore
If your app wins a 30% share of that	about ₹2.5 lakh crore of payments a year flowing through you
If you could charge merchants 0.3% on the merchant half of it	roughly ₹300–400 crore of annual revenue

Stop on that last line and think about it properly, because it is the uncomfortable one. ₹300 to ₹400 crore is a real business. Now set it against what it costs to acquire a hundred million users, run the infrastructure behind billions of transactions, and compete with three well-funded rivals doing exactly the same thing.

A student who does this arithmetic honestly in 2016 arrives somewhere awkward: the volume is enormous and the revenue is not obviously there. Hold that thought — it is the most important thing in this case, and section 6 is about what happened to it.

6 What Actually Happened, Including the Part That Went Wrong

Now check the 2016 estimate against reality, which is an exercise almost nobody performs and everybody should.

	THE 2016 ESTIMATE	WHAT HAPPENED BY FY2024
Payments across the whole system, per year	roughly ₹8.4 lakh crore	Monthly value running at roughly ₹20 lakh crore — annualised, in the region of ₹200 lakh crore
Transactions	ten a month per user	Around 131 billion UPI transactions in a single year
PhonePe's share	an aggressive 30%	Roughly 48% of all UPI transactions in India

The whole market came in more than twenty times larger than the careful estimate a diligent analyst would have produced in 2016. Every single assumption in section 5 was defensible at the time, and every one of them turned out to be low: more users, far more transactions per user, and a collapse in the size of a typical transaction as UPI turned from a way to send money to a friend into a way to buy a ₹20 packet of biscuits.

And now the uncomfortable half, which is the part of this story that rarely gets told in classrooms.

The revenue problem in section 5 was real, and it got worse rather than better. From January 2020 the government mandated a zero merchant discount rate on UPI and RuPay transactions — meaning the payment itself, the thing PhonePe now does billions of times a month, generates no direct fee at all. Consider what that does to the 2016 model: the ₹300 to ₹400 crore revenue line, already too thin, goes to approximately zero.

A company can own close to half of the largest payment network on the planet and earn nothing from the payments.

PhonePe's answer took seven years to find. If a hundred crore people open your app several times a week, you are no longer only a payments company — you are the most-visited financial storefront in India. The company built revenue on top of that traffic: insurance distribution, lending partnerships, mutual funds and stockbroking, merchant services including the soundbox device that announces payments aloud, and advertising.

PHONEPE REVENUE	FIGURE
FY23	approximately ₹2,914 crore
FY24	approximately ₹5,064 crore, with the company reporting an adjusted profit for the first time

The market was as large as they thought it would be. The business model was not the one the market implied.

7 Two Counter-Cases: Right Size, Wrong Answer

PhonePe read a market correctly. Here are two Indian businesses that also read a market as enormous, and were also right about the size, and still lost.

The wallets of 2012. Airtel Money and its contemporaries were pointed at exactly the market PhonePe later won. Their reasoning was identical: cash is inefficient, phones are everywhere, this must change. They were not wrong about any of it. Smartphones were too few, data was too expensive, and, most decisively, no interoperable rail existed — so each wallet could only reach its own users. Every ingredient PhonePe depended on arrived after these companies had already spent their money.

Paytm Mall. Launched in 2017 into Indian retail — a market of many lakh crore, unarguably enormous and correctly measured — backed by serious capital including from Alibaba, and attached to one of the most recognised consumer brands in the country. It never established a meaningful share against Flipkart and Amazon. Its value of goods sold fell rather than grew, and the business was effectively wound down and sold off by 2023. Two entrenched competitors had already built the supplier relationships, the logistics and the customer habit.

Three companies, then. All three looked at a market that really was enormous. One of them now handles roughly half of India's UPI transactions; one spent its money four years too early; one had the money, the brand and the category and could not take a piece of it.

8 The Question This Case Sets Up

In 2016, three men produced a number for a market that did not yet exist, and the number was wrong by a factor of twenty — in their favour. The revenue line they built on top of it was also wrong, in the other direction, and nearly fatally so.

Both errors came out of the same short list of assumptions in section 5. Some of those lines could have been checked in an afternoon in 2016. Others could not be checked at all, by anybody, because the behaviour they described had not happened yet.

So bring an answer to this. Go back to section 5 and look at the four or five lines that build the number. **If you could only research one of them properly — one line, one week of work — which would you pick, and why that one rather than the others?**

That question, asked about any business, is most of what the next session teaches. It is also, as you will discover in the first ten minutes of class, a question you can answer about the chai tapris outside this building.

BEFORE YOU COME TO CLASS

HOW TO USE THIS PAGE

Five questions. Question 1 needs paper and about three minutes — do not skip it, the class opens with it. The rest are a line or two each. Nothing here is collected or graded, but you will be asked for your answers in the room, and a wrong number you can explain is worth far more than no number at all.

There are no model answers at the back of this document. That is deliberate.

1. **The chai tapris outside this building. How much do they all earn in a year, together?** Work it out on paper before class and bring your working, not just the answer. Write down every number you used and, next to each one, mark whether you counted it, estimated it, or guessed it. Expect the room's answers to differ by ten times — that is the interesting part.

2. **Which of the three markets in section 4 do you think PhonePe was counting?** Commit to one, and give the single strongest argument against your own choice.

3. **Section 5 builds a number out of four or five lines.** Take any one of them and halve it, then double it. Which line, when you change it, moves the final answer the most? Bring the line and the two numbers it produces.

4. **PhonePe handles close to half of India's UPI transactions and earns nothing on the payments themselves.** Name one other thing you use every day that is free to you. Who pays for it, and what are they buying? Be specific — 'ads' is not yet an answer.

5. **Pick a small business you can actually see from where you live** — a laundry, a gym, a tiffin service, a printing shop. Estimate what it earns in a year using no more than five lines of arithmetic. Then write down which one of your five lines you are least sure about, and how you could check it this week without spending any money.

Bring your five answers, your working for question 1, and one number in this case that you think is doing more work than it deserves. Several of them are.